Pre-Seed Funding Requirements

AVRAI - Comprehensive Cost Breakdown

Executive Summary

This document provides a comprehensive breakdown of all costs required to bring AVRAI to a testable state for pre-seed investors. Costs are organized by category with minimum and maximum estimates to provide a realistic funding range.

Total Funding Range:

- **Minimum:** $40,357
- **Maximum:** $114,599
- **Recommended Ask:** $50,000 - $75,000

1. Legal & Incorporation

Item	Description	Minimum	Maximum	Notes
LLC/Corporation Filing	State filing fees, registered agent	**$200**	**$500**	Varies by state
Operating Agreement/Bylaws	Legal document preparation	**$500**	**$1,500**	Attorney fees
EIN Registration	IRS Employer Identification Number	**$0**	**$0**	Free from IRS
Business License	Local business license (if required)	**$50**	**$300**	Varies by location
Registered Agent Service	Annual registered agent (if needed)	**$50**	**$300**	First year
Legal Consultation	Initial legal setup consultation	**$400**	**$1,000**	Attorney review
TOTAL		**$1,200**	**$3,900**	

Notes: *Minimum assumes DIY filing with minimal legal review. Maximum assumes full attorney-assisted setup. Operating agreement critical for multi-founder equity splits.*

2. Intellectual Property (Patents)

Provisional Patents (Priority Establishment)

Item	Description	Minimum	Maximum	Notes
Provisional Patent #1	AI2AI Personality Learning System	**$2,000**	**$5,000**	Core technology
Provisional Patent #2	Quantum-Inspired Matching Algorithm	**$2,000**	**$5,000**	Matching system
Provisional Patent #3	Offline-First Mesh Network Protocol	**$2,000**	**$5,000**	Network architecture
Provisional Patent #4	Knot Theory Group Dynamics	**$2,000**	**$5,000**	Group matching
Provisional Patent #5	Privacy-Preserving Data Architecture	**$2,000**	**$5,000**	Privacy system
Patent Attorney Consultation	Strategy and prior art search	**$2,000**	**$5,000**	Initial consultation
Prior Art Search	Comprehensive prior art research	**$1,000**	**$3,000**	Per patent (optional)
TOTAL (Provisional Patents)		**$13,000**	**$33,000**	

Recommendation for Pre-Seed

- **File Provisional Patents First:** $13,000-$33,000
- Establishes priority date (12 months to file non-provisional)
- Lower initial cost while securing IP position
- Convert to non-provisional after seed round or revenue

Non-Provisional Patents (Full Filing - Optional for Pre-Seed)

Component	Small Entity	Large Entity	Notes
USPTO Filing Fees (per patent)	**$1,040**	**$2,080**	Includes filing, search, examination
Attorney Fees (per patent)	**$6,000**	**$21,000**	Claims drafting, prosecution, responses
Total per Patent	**$7,040**	**$23,080**	
Total for 5 Patents	**$35,200**	**$115,400**	

3. Hardware & Development Equipment

Development Machines

Item	Description	Minimum	Maximum	Notes
MacBook Pro (M3/M4)	Primary development machine (iOS builds require Mac)	**$1,999**	**$3,499**	14" base model to 16" maxed
Mac Mini (Backup/CI)	Secondary build machine or CI server	**$599**	**$1,299**	M2 base to M2 Pro
External Monitor	27" 4K monitor for development	**$299**	**$799**	Single to dual monitor setup
Keyboard & Mouse	Quality input devices	**$50**	**$200**	Basic to premium
TOTAL (Development Machines)		**$2,947**	**$5,797**	

Testing Devices (iOS)

Item	Description	Minimum	Maximum	Notes
iPhone 15 Pro	Latest iOS testing device	**$999**	**$1,199**	128GB to 256GB
iPhone 14	Secondary iOS device (older OS testing)	**$699**	**$799**	Used/refurbished acceptable
iPad (Optional)	Tablet testing (if app supports)	**$329**	**$599**	iPad 10th gen to iPad Air
TOTAL (iOS Devices)		**$1,698**	**$2,597**	

Testing Devices (Android)

Item	Description	Minimum	Maximum	Notes
Google Pixel 8	Latest Android testing device	**$549**	**$699**	128GB to 256GB
Samsung Galaxy S24	Secondary Android (different manufacturer)	**$699**	**$899**	Used/refurbished acceptable

Item	Description	Minimum	Maximum	Notes
Budget Android Device	Low-end device testing (Android Go)	**$149**	**$299**	Essential for compatibility
TOTAL (Android Devices)		**$1,397**	**$1,897**	

Additional Hardware

Item	Description	Minimum	Maximum	Notes
USB-C Hub/Dock	Connectivity for MacBook	**$49**	**$199**	Basic to Thunderbolt 4
External Storage	Backup and large file storage	**$79**	**$299**	1TB SSD to 4TB
Cables & Accessories	Chargers, cables, adapters	**$50**	**$150**	Various connectors
TOTAL (Additional Hardware)		**$178**	**$648**	
TOTAL HARDWARE		**$6,220**	**$10,939**	

4. Software Subscriptions & Services

Development Tools

Item	Description	Minimum	Maximum	Notes
GitHub Pro	Private repositories, CI/CD	**$48/year**	**$48/year**	$4/month
JetBrains IDE	IntelliJ IDEA / Android Studio (if used)	**$0**	**$149/year**	Free community or paid
VS Code Extensions	Premium extensions (optional)	**$0**	**$100/year**	Mostly free
Design Tools	Figma Pro, Adobe Creative Suite	**$0**	**$600/year**	Free tier or full suite
TOTAL (Development Tools)		**$48**	**$897**	**Annual**

Project Management & Collaboration

Item	Description	Minimum	Maximum	Notes
Notion Pro	Documentation and project management	**$96/year**	**$96/year**	$8/month
Slack	Team communication (if team > 2)	**$0**	**$1,044/year**	Free tier or $87/user/month
Linear/Asana	Issue tracking (if not using GitHub)	**$0**	**$120/year**	Free tier or paid
TOTAL (PM & Collaboration)		**$96**	**$1,260**	**Annual**
TOTAL SOFTWARE SUBSCRIPTIONS		**$144**	**$2,157**	**Annual**

5. Cloud Infrastructure & APIs

Backend Services

Item	Description	Minimum	Maximum	Notes
Supabase	Backend-as-a-Service (database, auth, storage)	**$0**	**$300/year**	Free tier to Pro ($25/mo)
Firebase	Analytics, messaging, storage (Google)	**$0**	**$300/year**	Spark (free) to Blaze (pay-as-you-go)
TOTAL (Backend Services)		**$0**	**$600**	**Annual (free tier to paid)**

Google Cloud Services

Item	Description	Minimum	Maximum	Notes
Google Places API (New)	Place search and details	**$0**	**$2,400/year**	$200 free credit/month, then $17/1K requests
Google Maps SDK	Map display (Android/iOS)	**$0**	**$2,400/year**	$200 free credit/month, then $7/1K loads
Google OAuth	Google Sign-In	**$0**	**$0**	Free
Google Cloud Storage	File storage (if needed)	**$0**	**$600/year**	Free tier to paid
TOTAL (Google Services)		**$0**	**$5,400**	**Annual (free tier to moderate usage)**

Weather & Other APIs

Item	Description	Minimum	Maximum	Notes
OpenWeatherMap	Weather data API	**$0**	**$480/year**	Free tier (1K calls/day) to One Call API 3.0
Stripe	Payment processing	**2.9% + $0.30**	**2.9% + $0.30**	Per transaction (no monthly fee)

Item	Description	Minimum	Maximum	Notes
TOTAL (Other APIs)		**$0**	**$480**	**Annual**
TOTAL CLOUD & APIs		**$0**	**$6,480**	**Annual**

Notes: *Most services offer generous free tiers for development/testing. Costs scale with usage (users, API calls, storage). Estimate based on moderate testing usage (not production scale).*

6. Developer Accounts & App Store Fees

Item	Description	Minimum	Maximum	Notes
Apple Developer Account	Required for iOS app distribution	**$99/year**	**$99/year**	Annual fee
Google Play Developer Account	One-time registration fee	**$25**	**$25**	One-time (not annual)
TOTAL DEVELOPER ACCOUNTS		**$124**	**$124**	**First Year: $124, Subsequent: $99/year**

7. Security & Compliance

Security Audit

Item	Description	Minimum	Maximum	Notes
Cryptographic Security Audit	Code review, penetration testing, spec verification	**$15,000**	**$30,000**	Required before production
TOTAL (Security Audit)		**$15,000**	**$30,000**	**One-time**

Notes: *Critical for AI2AI encryption and Signal Protocol implementation. Required before handling real user data. Can be deferred to post-seed if not handling sensitive data initially.*

Compliance & Legal

Item	Description	Minimum	Maximum	Notes
Privacy Policy	Legal document preparation	**$500**	**$2,000**	Attorney-drafted
Terms of Service	Legal document preparation	**$500**	**$2,000**	Attorney-drafted
GDPR Compliance Review	Privacy compliance (if targeting EU)	**$0**	**$3,000**	Optional initially
TOTAL (Compliance)		**$1,000**	**$7,000**	
TOTAL SECURITY & COMPLIANCE		**$16,000**	**$37,000**	

8. Testing & QA Equipment

Item	Description	Minimum	Maximum	Notes
Additional iOS Device	Older iOS version testing	**$0**	**$499**	Used iPhone 12/13 (optional)
Additional Android Device	Different Android version	**$0**	**$399**	Used Pixel 6/7 (optional)
Firebase Test Lab	Automated device testing	**$0**	**$600/year**	Free tier to paid
TOTAL TESTING & QA		**$0**	**$1,498**	

9. Professional Services

Item	Description	Minimum	Maximum	Notes
Ongoing Legal Consultation	Quarterly legal review	**$0**	**$2,000**	Optional
Contract Review	Vendor contracts, NDAs	**$0**	**$1,500**	As needed
Accounting Software	QuickBooks, Xero	**$0**	**$300/year**	Free tier or paid
Bookkeeping Service	Monthly bookkeeping (if outsourced)	**$0**	**$2,400/year**	Optional
Logo Design	Professional logo (if not DIY)	**$0**	**$1,500**	Fiverr to professional agency
App Icon Design	App store icons (if not DIY)	**$0**	**$500**	Optional
TOTAL PROFESSIONAL SERVICES		**$0**	**$8,200**	

10. Summary Tables

Category Summary

Category	Minimum	Maximum	Notes
1. Legal & Incorporation	**$1,200**	**$3,900**	Business formation
2. Intellectual Property	**$13,000**	**$33,000**	Provisional patents (5)
3. Hardware & Equipment	**$6,220**	**$10,939**	Mac, phones, accessories
4. Software Subscriptions	**$144**	**$2,157**	Annual subscriptions
5. Cloud Infrastructure & APIs	**$0**	**$6,480**	Annual (mostly free tier)
6. Developer Accounts	**$124**	**$124**	Apple + Google Play
7. Security & Compliance	**$16,000**	**$37,000**	Security audit + legal docs
8. Testing & QA	**$0**	**$1,498**	Additional devices/services
9. Professional Services	**$0**	**$8,200**	Legal, accounting, design
SUBTOTAL	**$36,688**	**$104,298**	
10. Contingency (10%)	**$3,669**	**$10,430**	**10% buffer**
TOTAL FUNDING NEEDED	**$40,357**	**$114,728**	

11. Alternative Scenarios

Scenario A: Minimal Viable Testing (Bare Minimum)

Assumptions: Use existing Mac (if available), file only 2 provisional patents (core IP), defer security audit to post-seed, use only free tiers for cloud services, minimal legal setup.

Category	Amount
Legal & Incorporation	**$1,200**
Intellectual Property (2 patents)	**$5,200**
Hardware (phones only, assume Mac exists)	**$3,095**
Software Subscriptions	**$144**
Developer Accounts	**$124**
Privacy Policy & ToS (DIY templates)	**$200**
SUBTOTAL	**$9,963**
Contingency (10%)	**$996**
TOTAL	**$10,959**

Scenario B: Comprehensive Pre-Seed (Recommended)

Assumptions: Full provisional patent portfolio (5 patents), complete hardware setup (Mac + testing devices), basic security review (not full audit), paid tiers for critical services, professional legal setup.

Category	Amount
Legal & Incorporation	**$2,500**

Category	Amount
Intellectual Property (5 provisional)	**$25,000**
Hardware & Equipment	**$8,500**
Software Subscriptions	**$1,000**
Cloud Infrastructure & APIs	**$2,000**
Developer Accounts	**$124**
Security Review (basic, not full audit)	**$5,000**
Compliance (Privacy Policy + ToS)	**$1,500**
Testing & QA	**$500**
Professional Services	**$2,000**
SUBTOTAL	**$48,124**
Contingency (10%)	**$4,812**
TOTAL	**$52,936**

Scenario C: Full Production Ready (Maximum)

Assumptions: Full non-provisional patent filing (5 patents), complete hardware setup, full security audit, all paid service tiers, comprehensive legal and compliance.

Category	Amount
Legal & Incorporation	**$3,900**
Intellectual Property (5 non-provisional)	**$115,400**
Hardware & Equipment	**$10,939**
Software Subscriptions	**$2,157**
Cloud Infrastructure & APIs	**$6,480**
Developer Accounts	**$124**

Category	Amount
Security Audit (full)	**$30,000**
Compliance	**$7,000**
Testing & QA	**$1,498**
Professional Services	**$8,200**
SUBTOTAL	**$171,698**
Contingency (10%)	**$17,170**
TOTAL	**$188,868**

12. Funding Recommendations

Recommended Pre-Seed Ask: $50,000 - $75,000

Rationale:

- **Covers Scenario B (Comprehensive Pre-Seed):** $52,936
- **Provides buffer for unexpected costs:** Additional $17,064 - $22,064
- **Allows for 6-12 months of operations** while building MVP
- **Demonstrates realistic planning** to investors

Minimum Viable Pre-Seed: $25,000

Rationale:

- **Covers core essentials:**
  - Legal setup: $1,200
  - 2-3 provisional patents: $6,000 - $9,000
  - Testing devices: $3,000
  - Developer accounts: $124
  - Basic compliance: $1,000
  - Contingency: $2,500
- **Requires existing Mac** (saves $2,000 - $3,500)
- **Uses free tiers** for cloud services initially
- **Defer security audit** to post-seed

Maximum Pre-Seed (If Seeking Full IP Protection): $150,000

Rationale:

- **Covers Scenario C essentials:**

    - Full non-provisional patents: $115,400
    - Complete hardware: $10,939
    - Full security audit: $30,000
    - All other categories: $39,182
- **Establishes strong IP position** before seed round
- **Production-ready security** from day one
- **Demonstrates serious commitment** to investors

13. Cost Optimization Strategies

Immediate Cost Savings

1. **Use Existing Hardware:** If you already have a Mac, save $2,000 - $3,500
2. **File Provisional Patents First:** Save $22,200 - $82,400 vs. non-provisional
3. **Use Free Service Tiers:** Most services offer generous free tiers for development
4. **Defer Security Audit:** Can wait until post-seed or first revenue
5. **DIY Legal Documents:** Use templates for Privacy Policy/ToS initially (upgrade later)

Long-Term Cost Management

1. **Monitor API Usage:** Set up billing alerts to avoid surprise charges
2. **Optimize Cloud Costs:** Use caching, CDN, and efficient queries
3. **Batch Patent Filings:** Some attorneys offer discounts for multiple patents
4. **Negotiate Service Contracts:** Annual plans often cheaper than monthly
5. **Consider Small Entity Status:** 50% discount on USPTO fees if eligible

14. Payment Timeline

Immediate (Month 1)

Item	Amount	Priority
Legal & Incorporation	**$1,200 - $3,900**	High
Developer Accounts (Apple + Google)	**$124**	High
Testing Devices (iOS + Android)	**$3,095 - $4,494**	High
TOTAL Month 1	**$4,419 - $8,518**	

Short-Term (Months 2-3)

Item	Amount	Priority
Provisional Patents (5)	**$13,000 - $33,000**	High
Development Machine (if needed)	**$1,999 - $3,499**	Medium
Privacy Policy & ToS	**$1,000 - $4,000**	High
TOTAL Months 2-3	**$15,999 - $40,499**	

Medium-Term (Months 4-6)

Item	Amount	Priority
Security Audit	**$15,000 - $30,000**	Medium (can defer)
Additional Hardware	**$178 - $648**	Low
Software Subscriptions	**$144 - $2,157**	Low
Cloud Services (if needed)	**$0 - $6,480**	Low (free tier initially)
TOTAL Months 4-6	**$15,322 - $39,285**	

Last Updated: January 26, 2026

Status: Ready for Review

Next Review: Before investor presentation